Study hours	Attendance y
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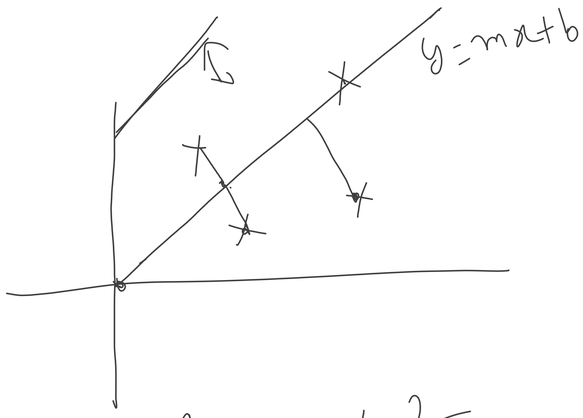


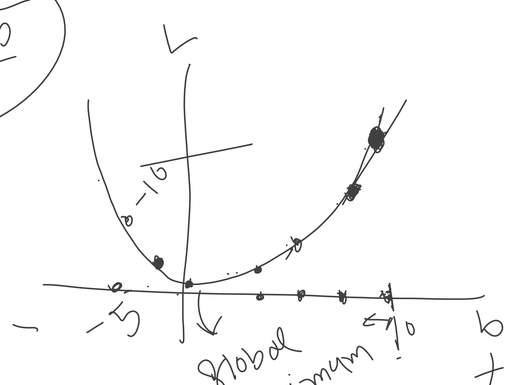
$$L = \sum_{i=1}^{4} (y - \hat{y})^2$$

$$L = \sum_{i=1}^{4} (y - mx_i - b)^2$$

$m = 10$

$$L(m, b) = \sum_{i=1}^{4} (y - 10x_i - b)^2$$

$L \to b$



+ve Slope
(-)ve [illegible] zone

(-)ve Slope
[illegible] (+) [illegible] zone

Slope +ve

$b_{new} = b_{old} - Slope \rightarrow$ $b = 10$, $Slope = 5$

$b = 10 - 5 = 5$

Slope -ve

$b_{new} = b_{old} + Slope$

$\Rightarrow b_{new} = b_{old} - Slope$

$\Rightarrow b_{new} = b_{old} - \eta \times Slope$

$\Rightarrow \boxed{b_{new} = b_{old} - \eta \times \frac{dL}{db}}$

$$L = \sum_{i=1}^{4} (y - 10 \cdot x_i - b)^2$$

$$\frac{dL}{db} = 2 \sum_{i=1}^{4} (y - 10 \cdot x_i - b) \cdot (-1)$$

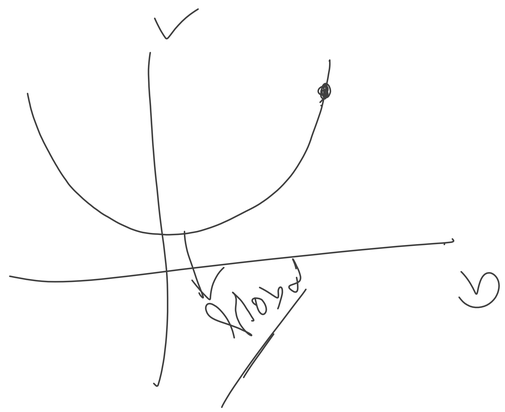
$$= -2 \sum_{i=1}^{4} (y - 10x_i - b)$$

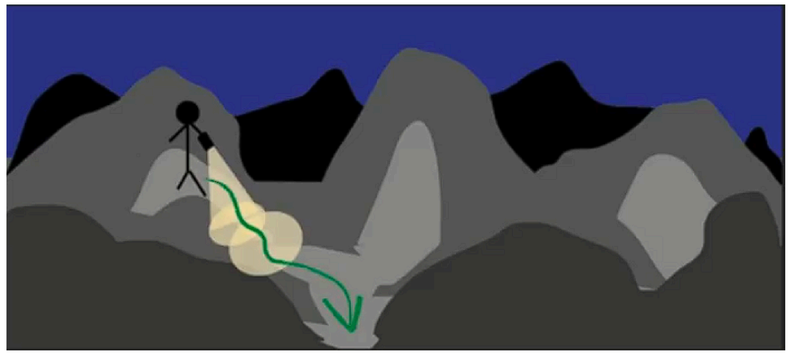
$\Rightarrow \boxed{b = 0}$

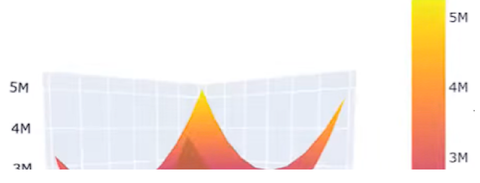
$$= \boxed{-2 \sum_{i=1}^{4} (y - 10x_i - 0)}$$

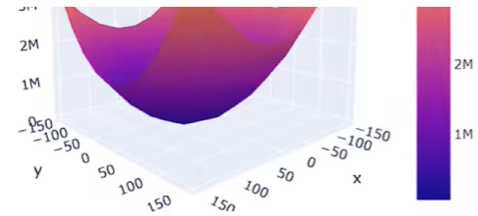
$\boxed{b_{new} = b_{old} - \eta \times \underline{slope}}$

$(\underline{m}, \underline{b})$









(m, b)

$$L = \sum_{i=1}^{4} (y - mx_i - b)^2$$

$$b_{new} = b_{old} - \eta \times \text{Slope}$$

$m = 1, b = 0$

Partial derivative

$$\frac{\partial L}{\partial b} = -2\sum (y - mx_i - b)$$

$$= -2\sum (y - mx_i - b) \times (-x_i)$$

$$\frac{\partial L}{\partial m} = -2\sum$$

$$m_{new} = m_{old} - \eta \times \text{slope}$$

$$\frac{\partial L}{\partial m} = -2\sum(y - mx_i - b) \times x_i$$